

數位電路實驗 Final team9

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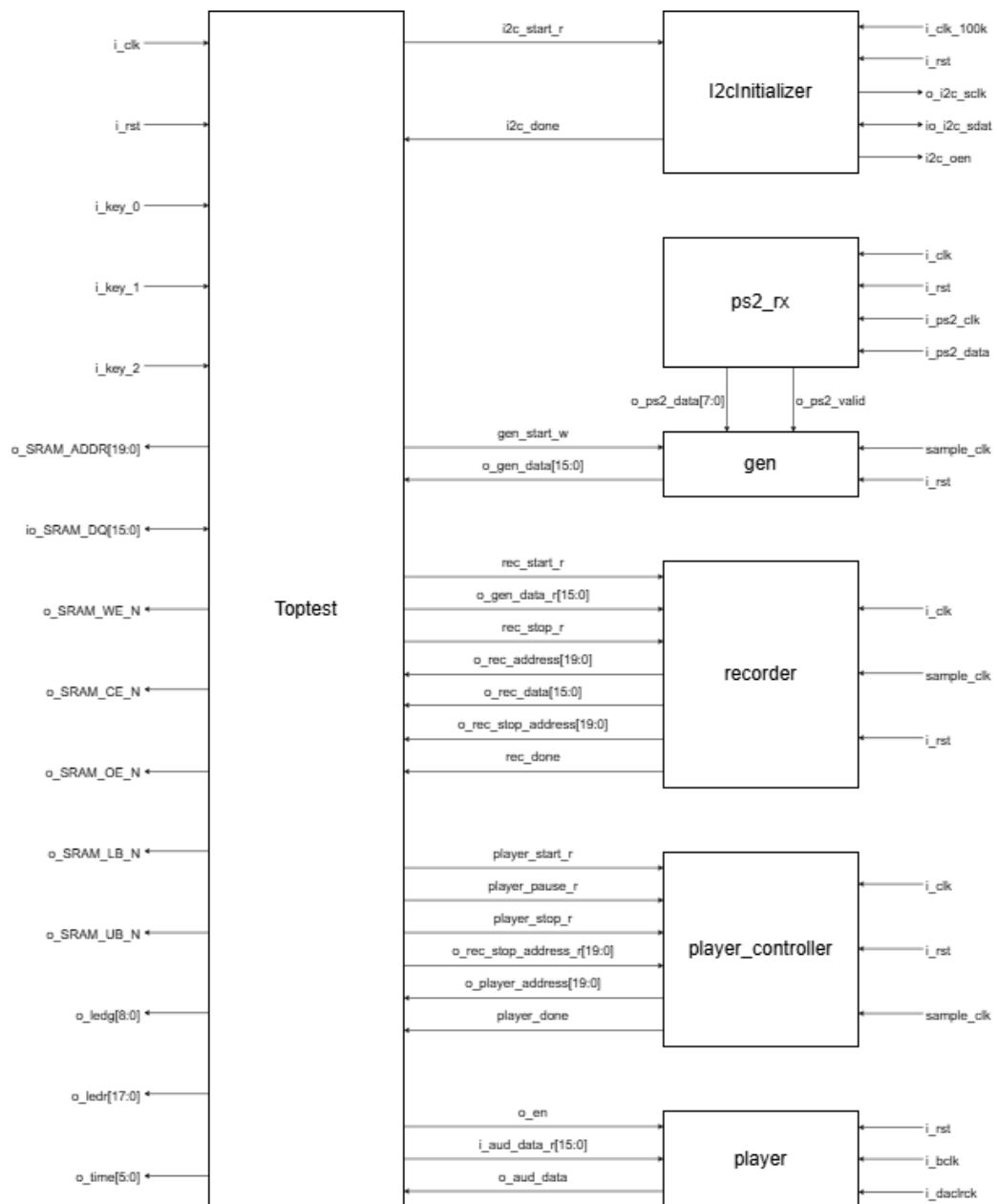
1. File Structure

```
team09_final
|- team09_final_report
|- src
|  |- DE2_115
|     |- DE2_115.qsf
|     |- DE2_115.sdc
|     |- DE2_115.sv
|     |- Debounce.sv
|     |- SevenHexDecoder.sv
|  |- Toptest.sv
|  |- I2cInitializer.sv
|  |- ps2_rx.sv
|  |- gen.sv
|  |- recorder.sv
|  |- player_controller.sv
|  |- player.sv
```

Module	Explanation
DE2_115	Connect FPGA with the outer equipment
Toptest	The top module
I2cInitializer	Initialize I2c
ps2_rx	Fetch the effective data generated by ps2 keyboard
gen	Generate wave
recorder	Control the recorder
player_controller	Controll the player
player	Control the output, connected to I2S

2. System Architecture

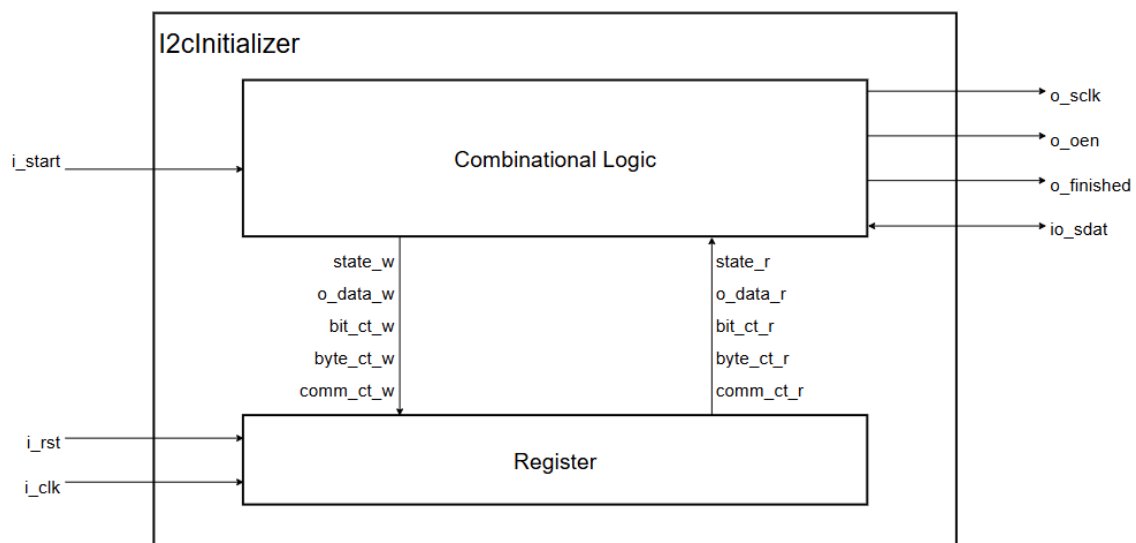
a. Toptest



Input/Output	Explanation
<i>i_clk</i>	50 MHz main clock
<i>i_rst</i>	Reset
<i>i_key_0</i>	Signal from FPGA key0. Control recorder/player to start
<i>i_key_1</i>	Signal from FPGA key1. Control player to pause
<i>i_key_2</i>	Signal from FPGA key2. Control recorder/player to stop
<i>i_clk_100k</i>	100 kHz clock. Use for I2c to initialize
<i>o_i2c_sclk</i>	Output clock of I2c

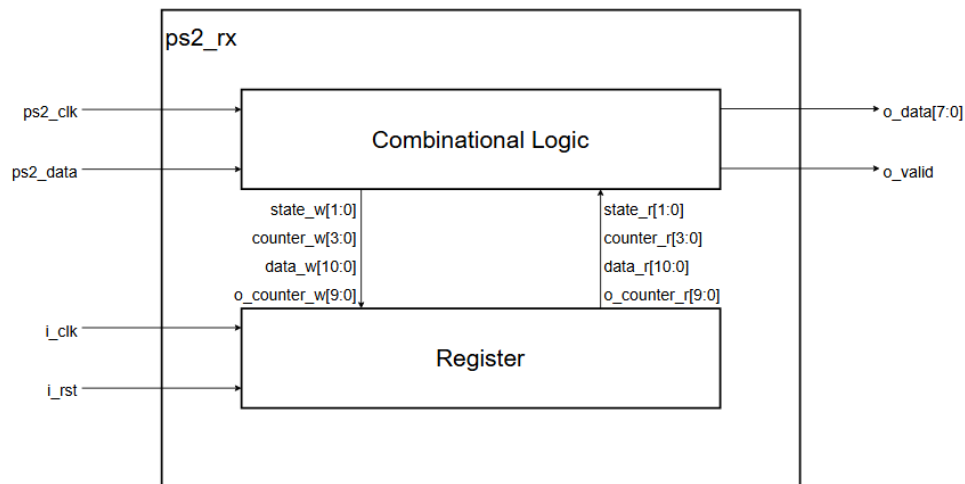
io_i2c_sdat	The output data and the input data from the slave
i_ps2_clk	10 - 16kHz clock generated by ps2 keyboard
i_ps2_data	Data transmitted from ps2 keyboard
sample_clk	50 kHz sampling clock
o_SRAM_ADDR[19:0]	Address of SRAM, tell SRAM which address need to be handled
io_SRAM_DQ[15:0]	Data read from/write to SRAM
o_SRAM_WE_N	Write enable, tell SRAM whether it is available to write
o_SRAM_CE_N	Chip enable, tell SRAM whether the chip is available
o_SRAM_OE_N	Output enable, tell SRAM whether it is available to output data
o_SRAM_LB_N	Lower byte enable, tell SRAM whether to handle SRAM_DQ[7:0]
o_SRAM_UB_N	Upper byte enable, tell SRAM whether to handle SRAM_DQ[15:8]
i_bclk	1.6 MHz bit clock, transmit the data bit
i_daclrck	50 kHz clock, the frequency of sound track changing
o_aud_data	The output to the speaker
o_ledg[8:0]	Green LED light
o_ledr[17:0]	Red LED light
o_time[5:0]	Record the time

b. I2cInitializer



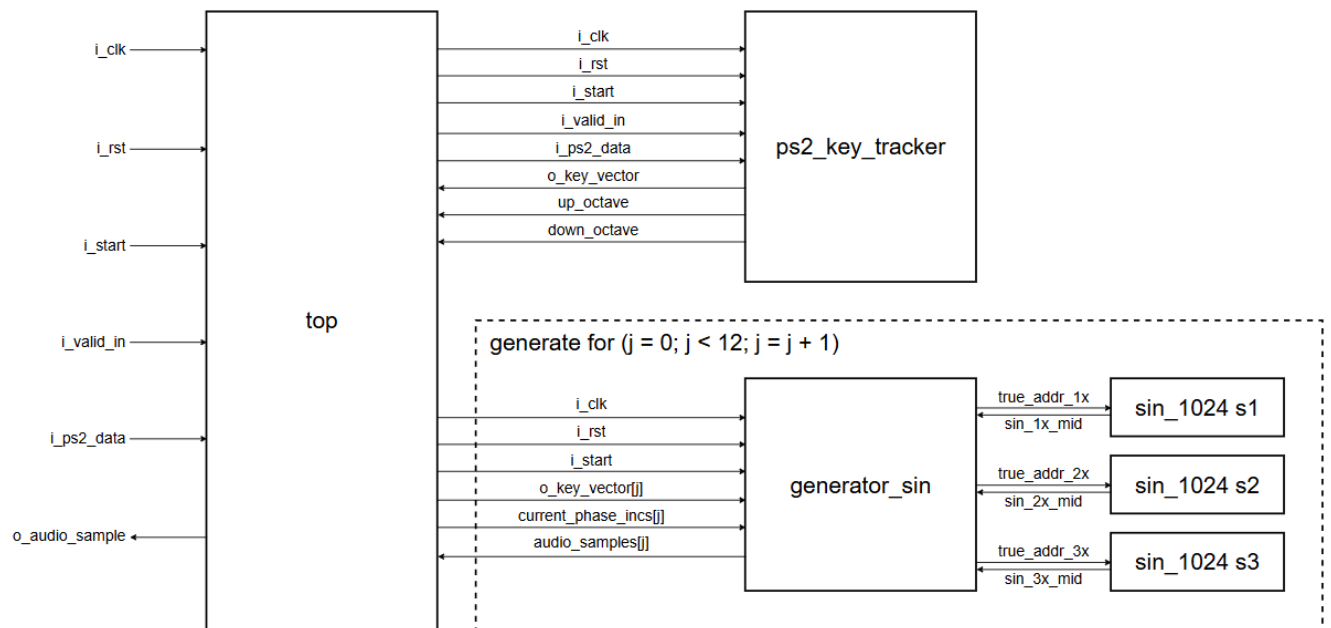
Input/Output	Explanation
i_rst	Reset
i_clk	100 kHz clock
i_start	Start the initializer.
o_sclk	The output clk of I2C.
o_oen	1 means io_sdat is output; 0 means io_sdat is input.
o_finished	Initializing is finished.
io_sdat	The output data and the input data from the slave.

c. ps2_rx

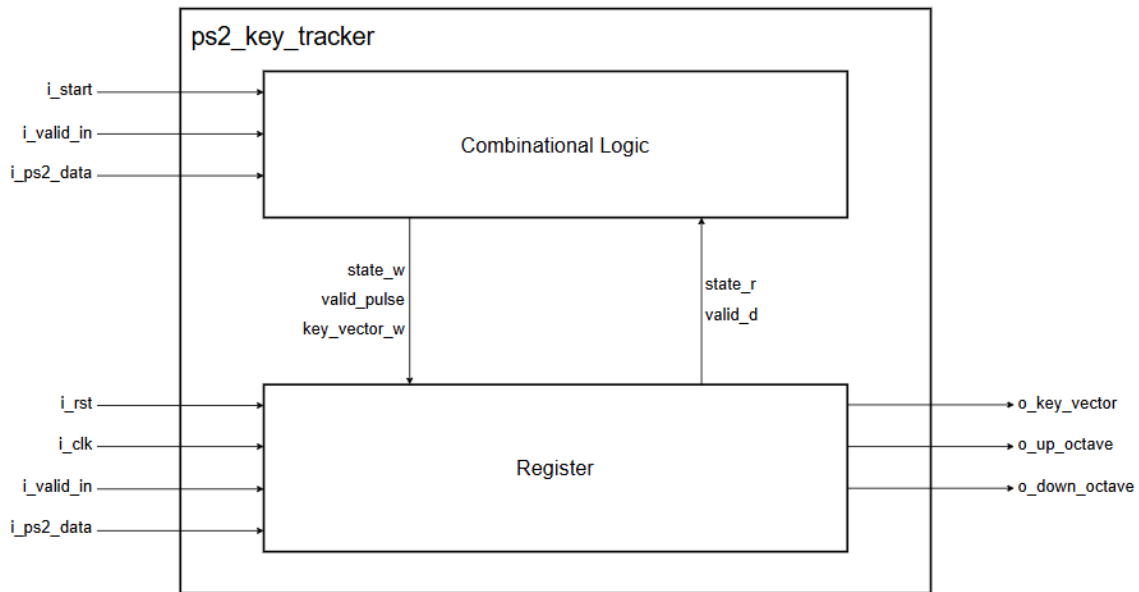


Input/Output	Explanation
i_clk	50 MHz clock
i_rst	Reset
ps2_clk	Clock generated by the ps2 keyboard. 10 - 16 kHz.
ps2_data	Data transmitted by the ps2 keyboard. Once a bit.
o_data[7:0]	Effective 8 bits data
o_valid	Check whether the output data is valid

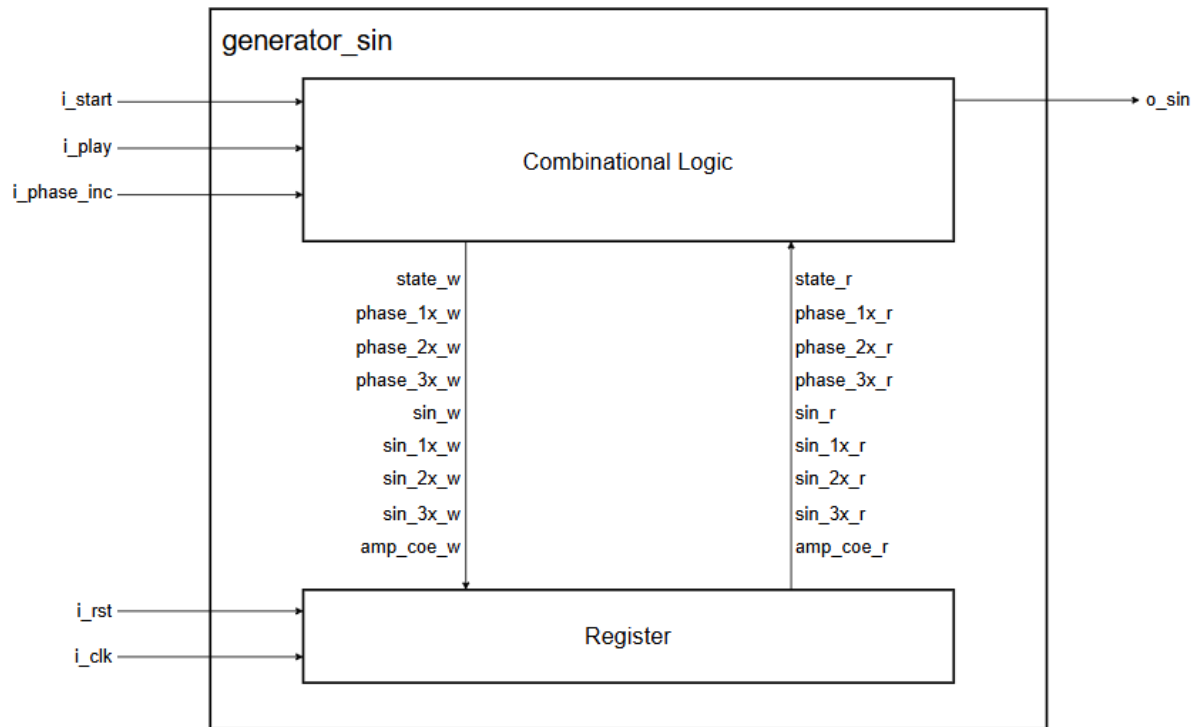
d. gen



Input/Output	Explanation
<i>i_clk</i>	50 kHz clock
<i>i_rst</i>	Reset
<i>i_start</i>	Start the ps2_key_tracker & generator_sin
<i>i_valid_in</i>	Indicates whether <i>i_ps2_data</i> is valid
<i>i_ps2_data</i>	Keyboard scan code
<i>o_audio_sample</i>	Output sound wave waveform (sum)

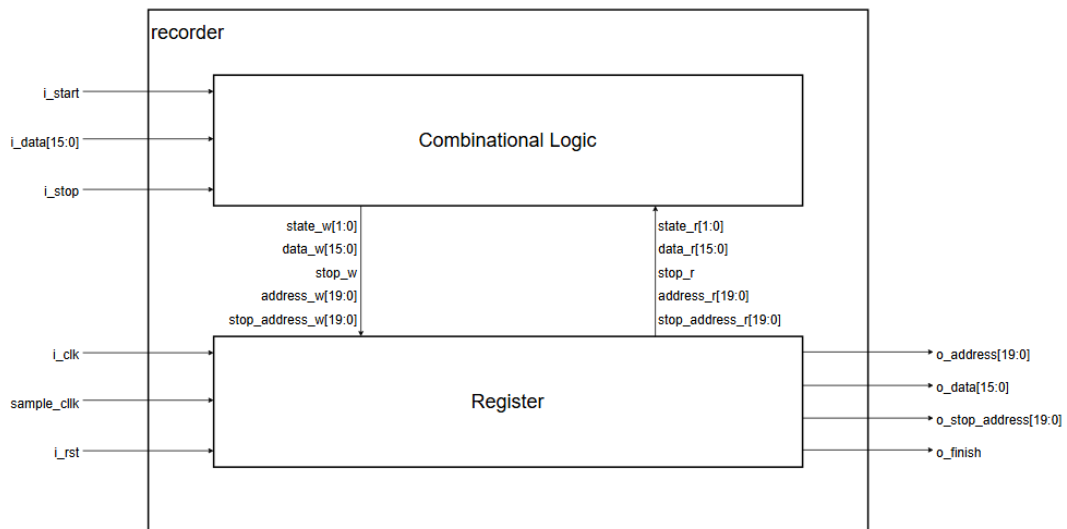


Input/Output	Explanation
<i>i_clk</i>	50 kHz clock
<i>i_rst</i>	Reset
<i>i_start</i>	Start
<i>i_valid_in</i>	Indicates whether <i>i_ps2_data</i> is valid
<i>i_ps2_data</i>	Keyboard scan code
<i>o_key_vector</i>	Each element in the vector indicates whether its corresponding key is being pressed.
<i>o_up_octave</i>	up an octave
<i>o_down_octave</i>	down an octave



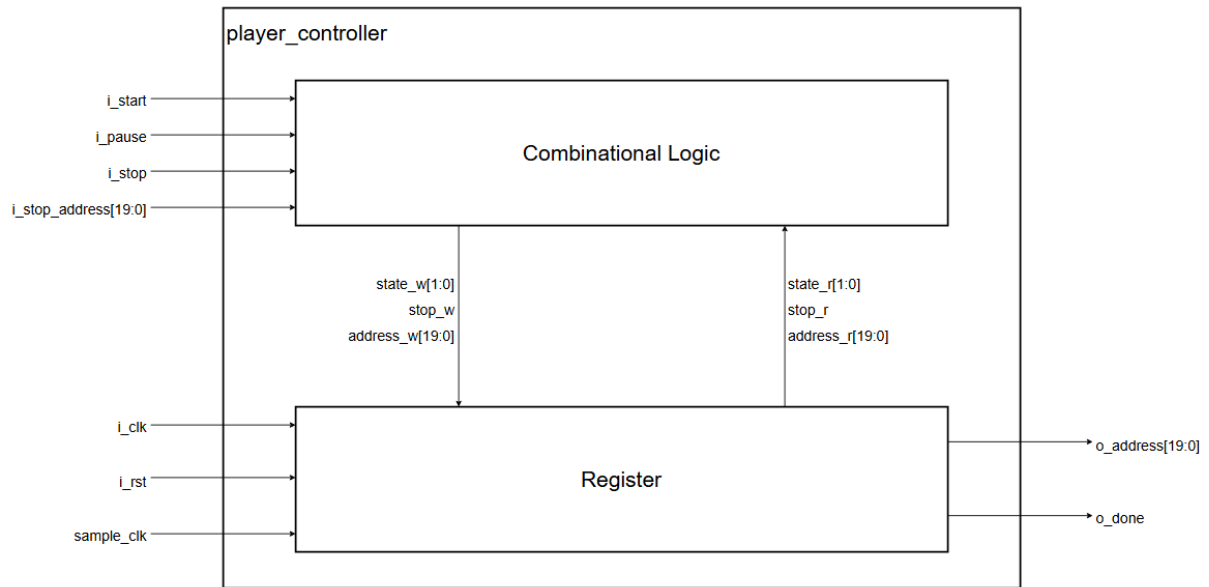
Input/Output	Explanation
i_clk	50 kHz clock
i_rst	Reset
i_start	Start
i_play	Indicates whether the corresponding key is being pressed.
i_phase_inc	the phase increase (proportional to the output frequency)
o_sin	Output sound wave waveform (single)

e. recorder



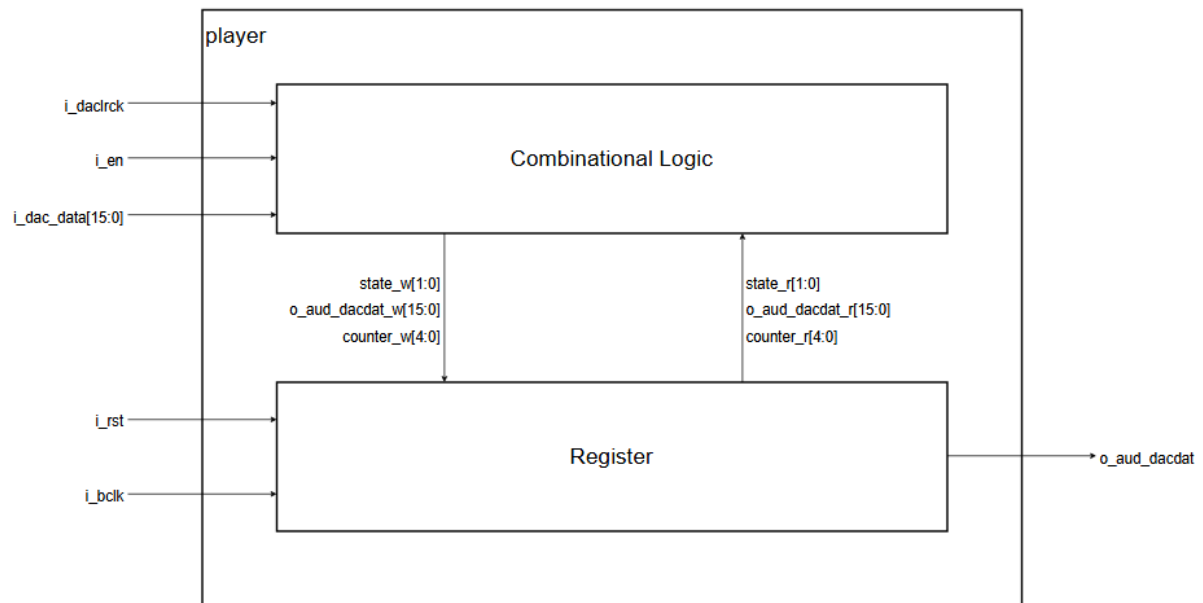
Input/Output	Explanation
i_clk	50 MHz main clock
sample_clk	50 kHz sampling clock
i_rst	Reset
i_start	Start to record
i_data[15:0]	Input data, need to be recorded
i_stop	Stop to record
o_address[19:0]	The address needs to record in
o_data[15:0]	The data needs to be record
o_stop_address[19:0]	The last address to be recorded
o_finish	Check whether the recording is finished

f. player_controller



Input/Output	Explanation
i_clk	50 MHz main clock
i_rst	Reset
sample_clk	50 kHz sampling clock
i_start	Start to play the audio
i_pause	Pause the audio
i_stop	Stop to play the audio
i_stop_address[19:0]	The last address stored by the recorder
o_address[19:0]	The address needs to be read
o_done	Check whether the the player is done

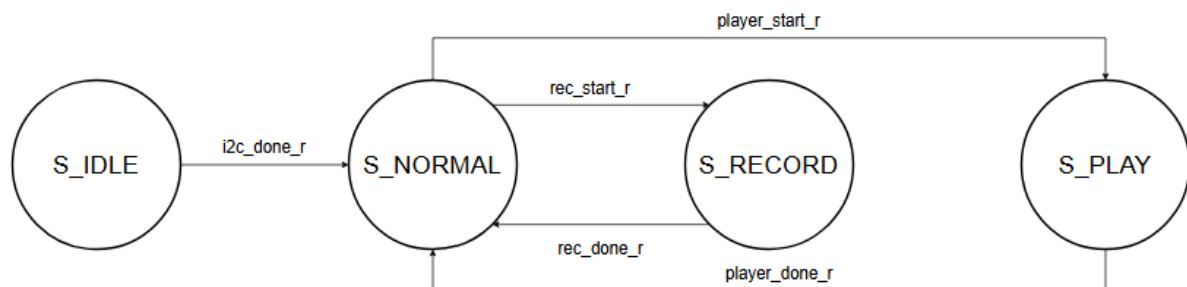
g. player



Input/Output	Explanation
i_rst	Reset
i_bclk	1.6 MHz bit clock
i_daclrck	50 kHz clock
i_en	Check whether to output
i_dac_data[15:0]	The data needs to be outputted
o_aud_dacdat	The output data

3. State Diagram & Explanation

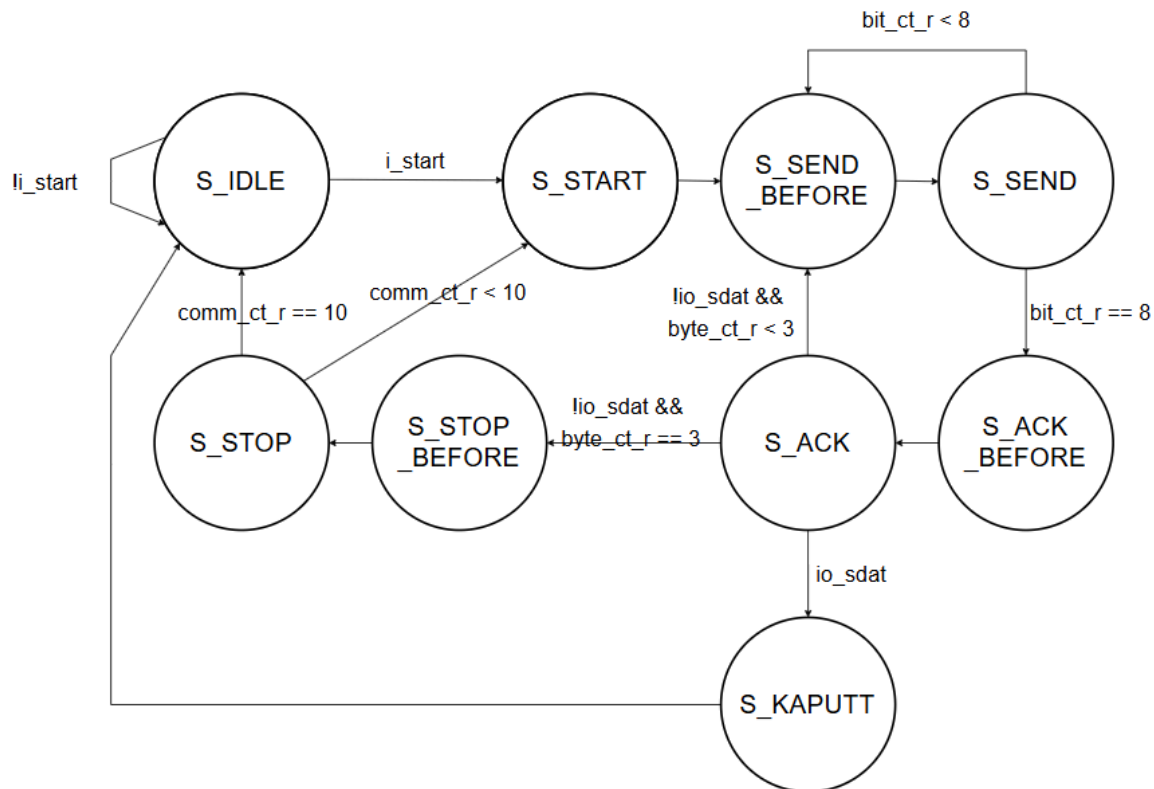
a. Topptest



State	Explanation
S_IDLE	Wait for I2c to finish initializing
S_NORMAL	The normal mode, press the keyboard to output audio

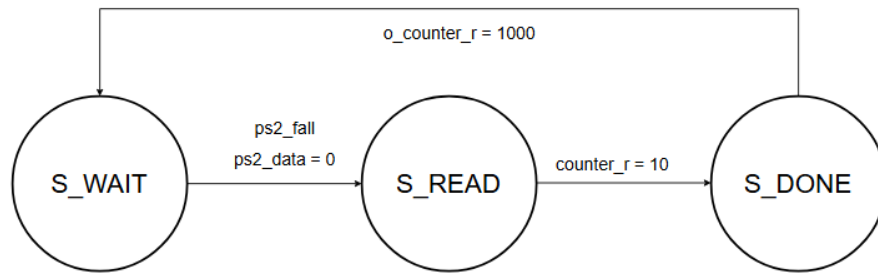
S_RECORD	Record mode, the audio will still be outputted, but the audio will be simultaneously stored in SRAM
S_PLAY	Play mode, start to output the data in SRAM

b. I2CInitializer



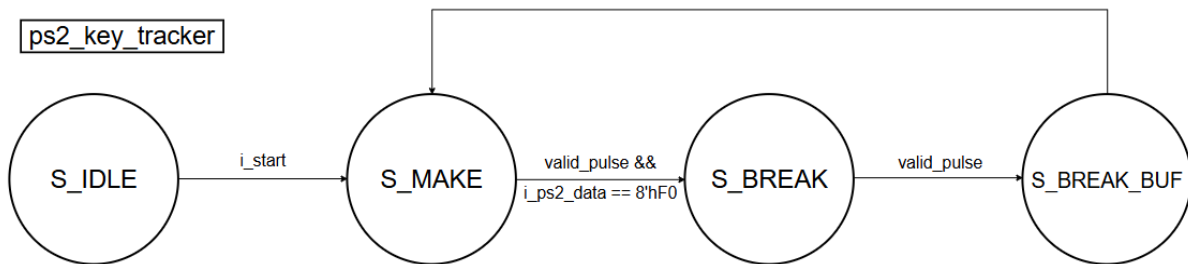
State	Explanation
S_IDLE	Wait for i_start.
S_START	Generates the I2C Start condition by pulling the data line (SDA) low.
S_SEND_BEFORE	Pulls the clock (SCLK) low to prepare the next bit for transmission.
S_SEND	Transmits a single bit of data onto the bus while the clock is high.
S_ACK_BEFORE	Releases the data line (High-Z) to allow the slave device to send an acknowledgment.
S_ACK	Samples the data line to check if the slave device sent an Acknowledgment (ACK).
S_STOP_BEFORE	Pulls the data line low to prepare for the I2C Stop condition.
S_STOP	Completes the current packet and loads the configuration data for the next register.
S_KAPUTT	Handles transmission errors (NACK) by aborting the operation and resetting.

c. ps2_rx

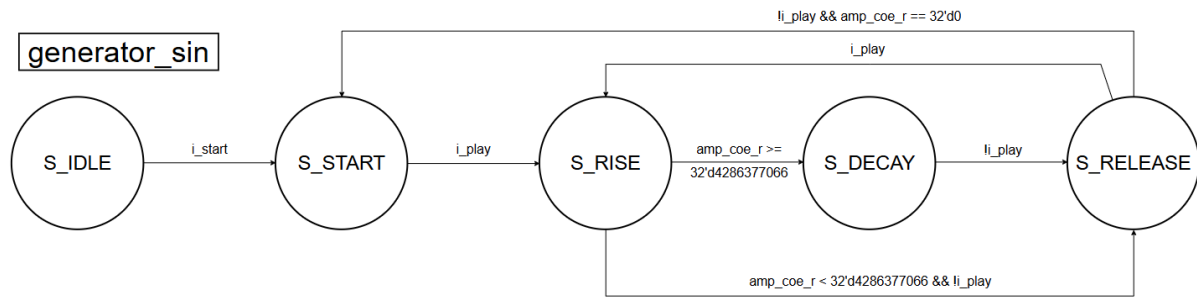


State	Explanation
S_WAIT	Wait for ps2 keyboard to start transmitting data
S_READ	Read and fetch the effective data
S_DONE	When ps2 finishing transmitting data, output o_data and o_valid

d. gen

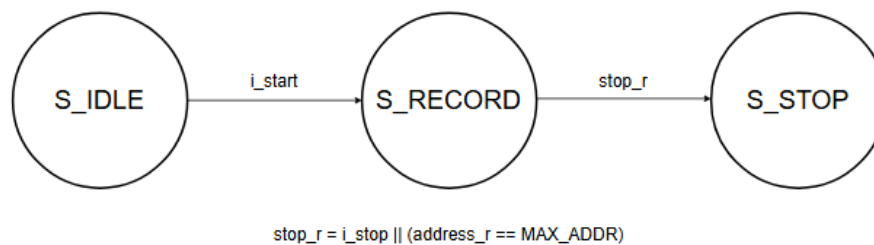


State	Explanation
S_IDLE	Wait for i_start.
S_MAKE	Make mode. The input is the key being pressed.
S_BREAK	Break mode. The next input is the key being released.
S_BREAK_BUF	Buffer state for S_BREAK



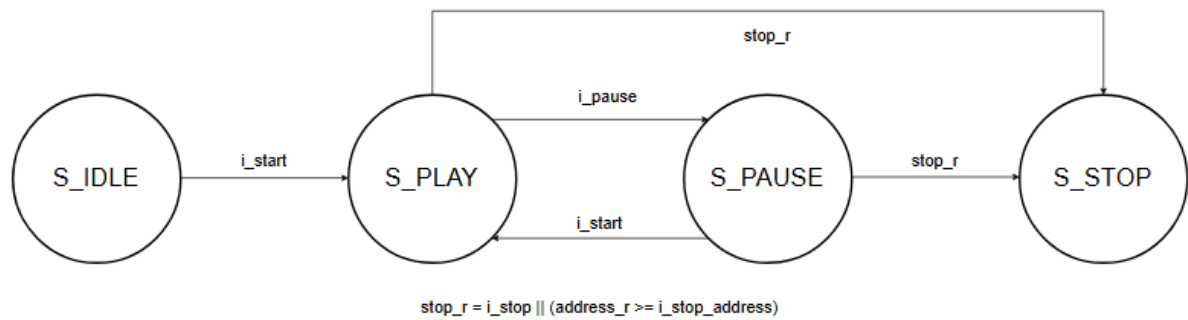
State	Explanation
S_IDLE	Wait for i_start.
S_START	Start to generate sin wave.
S_RISE	After being pressed, raise the amp_coe_r from 0 linearly.
S_DECAY	amp_coe_r decays exponentially
S_RELEASE	After being released, decrease the amp_coe_r linearly.

e. recorder



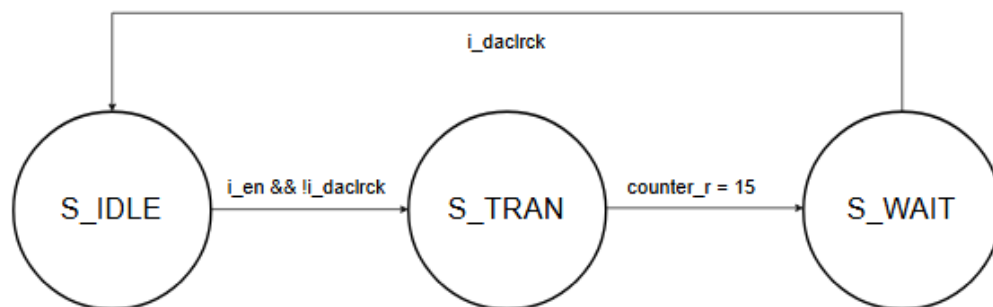
State	Explanation
S_IDLE	Wait for start
S_RECORD	Start to record, output the data and address need to be recorded in SRAM
S_STOP	Stop to record

f. player_controller



State	Explanation
S_IDLE	Wait for start
S_PLAY	Start to play the audio
S_PAUSE	Pause the audio
S_STOP	Stop to play the audio

g. player



State	Explanation
S_IDLE	Wait for i_en and the audio channel to be left channel
S_TRAN	Right channel, start to play for 16 bits
S_WAIT	Already write 16 bits, wait for the audio channel becomes right channel

4. Fitter Summary

Fitter Summary	
Fitter Status	Successful - Mon Dec 22 22:00:00 2025
Quartus II 64-Bit Version	15.0.0 Build 145 04/22/2015 SJ Full Version
Revision Name	DE2_115
Top-level Entity Name	DE2_115
Family	Cyclone IV E
Device	EP4CE115F29C7
Timing Models	Final
Total logic elements	55,015 / 114,480 (48 %)
Total combinational functions	52,446 / 114,480 (46 %)
Dedicated logic registers	5,536 / 114,480 (5 %)
Total registers	5536
Total pins	518 / 529 (98 %)
Total virtual pins	0
Total memory bits	115,712 / 3,981,312 (3 %)
Embedded Multiplier 9-bit elements	120 / 532 (23 %)
Total PLLs	1 / 4 (25 %)

5. Timing Analyzer

Table of Contents		Unconstrained Paths		
- Flow Non-Default Global Settings - Flow Elapsed Time - Flow OS Summary - Flow Log > Analysis & Synthesis > Fitter - Summary - Settings - Parallel Compilation - I/O Assignment Warnings - Netlist Optimizations - Ignored Assignments > Incremental Compilation Section - Pin-Out File > Resource Section > I/O Rules Section - Device Options - Operating Settings and Conditions - Messages - Suppressed Messages - Flow Messages - Flow Suppressed Messages > Assembler > TimeQuest Timing Analyzer - Summary - Parallel Compilation - SDC File List - Clocks > Slow 1200mV 85C Model > Slow 1200mV 0C Model > Fast 1200mV 0C Model - Multicorner Timing Analysis Summa > Multicorner Datasheet Report Sum > Advanced I/O Timing > Clock Transfers - Setup Transfers - Hold Transfers - Recovery Transfers - Removal Transfers - Report TCCS - Report RSKM - Unconstrained Paths - Messages				
		Property	Setup	Hold
1	Illegal Clocks	0	0	
2	Unconstrained Clocks	15	15	
3	Unconstrained Input Ports	32	32	
4	Unconstrained Input Port Paths	2879	2879	
5	Unconstrained Output Ports	69	69	
6	Unconstrained Output Port Paths	266	266	

6. 心得

這次是我們第一次接觸final project, 慢慢地看著自己的作品從無到有, 自發想開始, 到分析內部模組構造, 再到寫程式實作, 最後展出成品, 這些過程都給了我們非常有意義的經驗!

在這次的project中, 由於每個人主題不一, 沒有人可以協助我們, 遇到問題時只能慢慢地自己抽絲剝繭、上網查詢、詢問AI, 有時也會在電腦前面大吼, 開始崩潰, 雖然過程很艱苦, 但看到成品的那刻覺得一切都是值得的。

最後, 感謝串場演出的綠色椰子口味乖乖, 沒有這包我們可能做不出來(x